Abstract
The study of agriculture assumes that productivity fundamentally depends on appropriate classification of crops for improved decision-making and efficient resource management. In this regard, we analyze various supervised machine learning models to develop a crop classification system based on important environmental and soil parameters. We implement models in both R and Python and use Logit, Random Forest, Gradient Boosting, Support Vector Classifier, K-Klas Komsh, Decision Tree, and Support Vector Machine. Among all of these, the model that achieved the highest accuracy of 99.77% within R, surpassing 99.55% in Python, was Random Forest. The rest precision, recall, F1 and confusion matrix evaluation confirmed the performance of these models. The study is a proof of concept on applying machine learning in agriculture with suggestive directions for further work revolving around feature engineering and ensemble methods of learning.

I. INTRODUCTION
For efficient crop growing, assigning the right crop type, along with its expected yield, is crucial, therefore, machine learning (ML) has recently been a helpful component for calculating agricultural choices through the analysis of the environment and soil quality. Contempory methods of classification suffer from highly dimensional datasets. The intricate dependency relations among the controls are more efficiently resolved by ML techniques.

This research analyzes performance of several ML methods, to find the best model in terms of accuracy regarding crop classification and yield prediction. An integrated approach is suggested with soil data, crop data, weather data, and soil attribute data to improve classification precision. The effectiveness of further decision trees, support vector machines (SVM), ensemble learning, and deep learning methods will be measured for their efficacy in crop classification. Evaluation will be based on classification precision, accuracy, recall, and F1 score to provide comprehensive outcomes.

In addition, various XAI techniques will be used to make the model more interpretable and transparent. Environmental and soil factors affecting crop classification will be analyzed through feature importance, shapley addiititve explanation (SHAP) analyses, and local interpretable model agnostic explanations (LIME). With this information, farmers and other users of ML-based classification models will have more confidence in trusting these models.

The contributions of the study are listed below:

1. Steps Towards Using Machine Learning Tools for Crop Production: Implementation and analysis of several machine learning algorithms for classification and estimation of crop yields.

2. Engineering and Selection of features designed: Study of the most relevant variables that determine the accuracy of classification and estimation of yields.

3. Application of Theories on Interpretation and Explanation: Use of techniques of artificial intelligence to provide requisite explanation on the model output so that decision makers will have sufficient reasons to act on the predictions out of the model.

4. Generalization Across Agricultural Datasets: Estimation of model efficacy across various agricultural datasets for flexibility and strength analysis.

The rest of this document is organized as follows. In Section II, the proposed approach, consisting of data cleansing, feature extraction, and model design, is discussed. Section III contains the results of the experiments which include the comparison of various ML algorithms and their performance analysis. In the end, Section V of the report provides the conclusions of the research and discusses possible follow-up work in automating the analysis of crop types and estimating their yields using machine learning technologies.

II. Methodology

In this paper, I have developed and refined different machine learning models for predicting in agriculture. The procedure includes data cleansing, model definition, and hyperparameter estimation, followed by validation of the model through cross validation procedures.

1. Data Preprocessing

In order to prepare the input for the machine learning models, the models underwent the following steps for preprocessing.

- **Cleaning The Data:** Solving issues with absent values, deleting repeat entries, and making changes to entries with contradictions.
- **Feature Development:** Improving model performance with new features created by pre-existing domain knowledge.
- **Setting Normal Values:** Maintaining uniformity in data for those models that are adjusted to vertical proportions.
- **Splitting the Data:** Dividing data into parts and measuring the efficiency of solving the given problem by using distinct blocks of data.

2. Model Selection and Justification

In complex datasets several regression models are used and they include the following*:*

2.1 Decision Tree Regressor

Reasoning:

Non-linear relationships are captured and can be explained.

Optimization:

- Hyperparameters tuned include: minimum samples per split, minimum leaf size, tree depth.Optimized hyperparameters by Grid Search and Random Search.
- Performance generalization was ensured by 5-fold cross validation.

2.2 Random Forest Regressor

Reasoning:

Improves accuracy with multiple trees to predict unlike single decision trees.

Optimization:

- Key hyperparameters: number of trees, features per split, max depth, and min samples per split.
- Random and grid search were used for tuning.
- 5-fold cross validation guarantees accuracy and prevents overfitting.

2.3 K-Nearest Neighbors Regressor (KNNR)

Reasoning:

Useful for detecting local patterns and tendencies.

Optimization:

- Important hyperparameters: neighbor count, weighting approach, search method.
- Balance bias-variance trade-off using Grid Search and Random Search.
- Generalize performance using 5-fold cross validation.

2.4 Support Vector Regressor (SVR)

Reasoning:

Efficient in dealing with high dimensional number and capturing non-linear relations.

Optimization:

- Tuned parameters: ε , C(idual), kernel type (polynomial, RBF, linear), and γ (eps).
- Use random search and grid search for optimization.
- Use 5-fold cross-validation to control complexity vs generalization.

2.5 Extreme Gradient Boosting Regressor (XGBoostR)

Reasoning:

A boosting model set known for a particular capability of dominating large datasets with better accuracy and efficiency.

Optimization:

- Important hyperparameters: gamma, column sample, subsample, number of estimators, max_depth, and eta (learning rate).
- Optimal parameter selection with grid and random search.
- Generalization and robustness ensuring through 5-fold cross validation.

3. Model Evaluation

- **Performance Metrics**: Models were evaluated using key regression metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2).
- **Cross-Validation**: 5-fold cross-validation was applied to reduce bias and validate model reliability.
- **Comparison & Selection**: The best-performing model was selected based on evaluation metrics and computational efficiency.

By implementing these rigorous methodologies, the study aimed to develop an accurate and robust model for predicting crop yields based on relevant agricultural and environmental factors.

III. Experimental Results

3.1 Model Performance in Python

The performance of different machine learning models implemented in Python is summarized in the table below:

Model	Accuracy
Logistic Regression	95.23%
Random Forest	99.55%
Gradient Boosting	98.41%
SVC	97.95%
KNN	97.50%
Decision Tree	97.95%

3.2 Model Performance in R

The following table presents the accuracy of machine learning models implemented in R:

Model	Accuracy
Random Forest	99.77%
SVM	98.41%

3.3 Observations

- The Random Forest model built in R once again recorded the highest accuracy of 99.77%, more than the 99.55% attained in Python.
- The Python SVC and gradient boosting models each registered strong results in exceeding 97% accuracy.
- With hyperparameter tuning, the Python Decision Tree achieved an accuracy of 97.95%.
- Both models in Python and R showed that recall for the 'Rice' class was lower suggesting potential chances for errors in classification.

3.4 Hyperparameter Tuning for Decision Tree in Python

The Decision Tree model underwent hyperparameter tuning, yielding the following optimal settings:

- **Criterion**: Gini Impurity
- **Max Depth**: 20
- **Min Samples Leaf**: 2
- **Min Samples Split**: 11

These adjustments improved accuracy to **97.95%**, but the model was still outperformed by **Random Forest**.

3.5 Confusion Matrix Analysis

The confusion matrix indicated that most crops were classified correctly. However, some misclassifications occurred in the **'Jute'** and **'Rice'** classes, suggesting a need for **feature engineering** or **dataset balancing**.

IV. Results and Discussion

4.1 Overview of Model Performance

These set of studies aimed at applying machine learning models for crop classification and prediction of wheat yield. 7 classification models were attempted - Random Forest, Extra Trees, Logistic Regression, Decision Tree, K-Nearest Neighbors, Gaussian Naive Bayes and Support Vector Machine.

Additionally, for forecasting wheat yield, at least 5 different regression models were utilized along with a Decision Tree Regressor. To test the models in the most accurate way, the data was split into 70 percent as training data, and 30 percent as testing data.

4.2 Data Collection

- The dataset used for crop classification contained **2200 records** with **22 crop labels** (e.g., rice, wheat, cotton, bananas, apples).

- Features included **nitrogen concentration, temperature, soil pH, rainfall, humidity, phosphorus, and potassium content**.
- The dataset was sourced from **Kaggle**.

4.3 Classification Model Performance

- The **Random Forest Classifier** achieved the highest accuracy (**99.7%**), outperforming other models.
- **Extra Trees Classifier** and **Gaussian Naive Bayes** followed closely with **99.4% and 99.2% accuracy**, respectively.
- Logistic Regression, KNN, and SVM achieved **95.9%, 98.1%, and 98.9% accuracy**, respectively.

4.4 Feature Importance Analysis

Feature importance varied across different models:

- **Random Forest** prioritized **humidity (0.199)** and **rainfall (0.167)**.
- **Decision Tree** focused on **rainfall (0.263) and phosphorus (0.227)**.
- **Extra Trees Classifier** emphasized **humidity (0.178) and potassium (0.169)**.

These findings highlight the importance of environmental factors in crop classification.

4.5 Regression Model Performance for Wheat Yield Prediction

To predict **wheat yield**, five regression models were evaluated:

Model	RMSE	MAE	MSE
Support Vector Regressor	0.3236	0.1614	0.1047
K-Nearest Neighbors	Higher error rates		
Extreme Gradient Boosting	Higher error rates		
Random Forest	Higher error rates		
Decision Tree	Higher error rates		

- **Support Vector Regressor** outperformed other models with the lowest **RMSE (0.3236), MAE (0.1614), and MSE (0.1047)**.
- **KNN and Extreme Gradient Boosting** showed the highest error rates.

4.6 Cross-Validation and Model Interpretability

- **Cross-validation** was performed to ensure robustness.
- **LIME and SHAP** were used for model interpretability, showing how different features influenced predictions.
- Results indicate that **humidity and rainfall** are critical factors for crop classification and wheat yield prediction.

4.7 Conclusion

- **Random Forest** is the most effective classification model for **crop categorization**.
- **Support Vector Regressor** demonstrated superior performance in **wheat yield prediction**.
- **Feature selection and dataset balancing** play a crucial role in improving model accuracy.

These findings provide valuable insights for **agricultural planning and decision-making**.

V. Conclusion and Future Work

This research shows that machine learning models can improve crop classification tremendously. The model that I developed in R using the Random Forest algorithm outperformed all of the other models with an accuracy of 99.77%, beating the Random Forest model in Python that only achieved 99.55%. After hyperparameter tuning, Decision Tree also performed well, however, it was still lower than that of Random Forest model's. The results reflect that machine learning is another possible solution in agriculture provided that decisions made regarding crop classification are done with utmost precision and dependability.

Future work should focus on:

- **Feature Engineering**: Improving the selection of features and scaling them so the model performs better.
- **Augmentation of the Training Set:** Improving the training set with the use of generated synthetic data and balancing the training set to make the classification more robust.
- **Combining two or more classifiers for an Ensemble Learning model:** Where different models are trained with a purpose of improving the accuracy of classification.
- **Deep Learning Methods**: In these models neural networks, as well as their composition with deep learning, are the dominating type of model for more accurate classification of crops with the use of convolutional neural networks (CNNs) and recurrent neural networks (RNNs) for spatial and time series data.
- **Explainable AI (XAI):** Use of techniques from XAI to provide better qualitative conclusions from the model such as what soil pH and rainfall does to the predictions so that it is more understandable by the farmers.

- **Real-time Data Collection**: Utilizing IoT sensors for real-time data acquisition to improve predictive accuracy and support precision agriculture.

- **Scalability and Adaptability**: Testing the models across diverse geographical locations and integrating them into mobile applications to provide farmers with accessible and actionable insights.

The outcomes of this research can have significant practical implications for agricultural productivity and sustainability. By implementing these advancements, future research can further refine crop classification models, ultimately supporting data-driven agricultural practices and policy-making.
